

MedMemoryBench: Benchmarking Agent Memory in Personalized Healthcare

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 Code and Dataset: [AQ-MedAI/MedMemoryBench](https://github.com/AQ-MedAI/MedMemoryBench)

Abstract

The large-scale deployment of personalized healthcare agents demands memory mechanisms that are exceptionally precise, safe, and capable of long-term clinical tracking. However, existing benchmarks primarily focus on daily open-domain conversations, failing to capture the high-stakes complexity of real-world medical applications. Motivated by the stringent production requirements of an industry-leading health management agent serving tens of millions of active users, we introduce **MedMemoryBench**. We develop a human-agent collaborative pipeline to synthesize highly realistic, long-horizon medical trajectories based on clinically grounded, synthetic patient archetypes. This process yields a massive, expertly validated dataset comprising approximately 2,000 sessions and 16,000 interaction turns. Crucially, MedMemoryBench departs from traditional static evaluations by pioneering an *evaluate-while-constructing* streaming assessment protocol, which precisely mirrors dynamic memory accumulation in production environments. Furthermore, we formalize and systematically investigate the critical phenomenon of memory saturation, where sustained information influx actively degrades retrieval and reasoning robustness. Comprehensive benchmarking reveals severe bottlenecks in mainstream architectures, particularly concerning complex medical reasoning and noise resilience. By exposing these fundamental flaws, MedMemoryBench establishes a vital foundation for developing robust, production-ready medical agents. Our dataset and benchmark are publicly available at <https://github.com/AQ-MedAI/MedMemoryBench>.

1 Introduction

Driven by the rapid advancement of large language model (LLM) technologies, personalized healthcare agents are rapidly transitioning from proof-of-concept systems to large-scale deployment. For instance, Ant Group’s health management agent “Ant Afu” [1] currently serves tens of millions of active users, providing continuous assistance in chronic disease management, proactive health follow-ups, and medication guidance. In such long-horizon deployments, agent memory is not merely a conversational enhancer; it is the critical infrastructure required to maintain evolving patient profiles, medication histories, and strict safety constraints. Recent works have made notable progress in knowledge graph enhanced memory [7, 10, 25, 29], dedicated memory frameworks [5, 19, 22, 37], and reinforcement learning-optimized memory [33, 38–40], yet these approaches have been predominantly validated in general domains such as daily conversations, and remain insufficiently examined in large-scale, high-risk medical dialogue scenarios.

The medical domain imposes uniquely stringent requirements that shatter the fundamental assumptions of general-purpose memory systems. First, medical inquiries demand **exceptional precision**; semantically similar complaints, such as “diffuse abdominal pain” versus “right lower abdominal pain”, dictate fundamentally different diagnostic pathways, necessitating zero-tolerance retrieval accuracy. Second, agents must accurately track **long-horizon clinical evolution**, such as a patient’s fasting blood glucose shifting from 7.2 mmol/L to 9.8 mmol/L over a couple of days.

To evaluate the performance of existing methods under these challenges, targeted evaluation benchmarks are indispensable. However, existing benchmarks have been predominantly developed for general-purpose scenarios: LoCoMo [21] focuses on long-term understanding in multi-turn daily conversations, while other representative works address long-context processing [2, 13], reasoning ability [14, 34], or persona consistency alignment [15, 35], respectively. Although

*Work done during an internship at Ant Group.

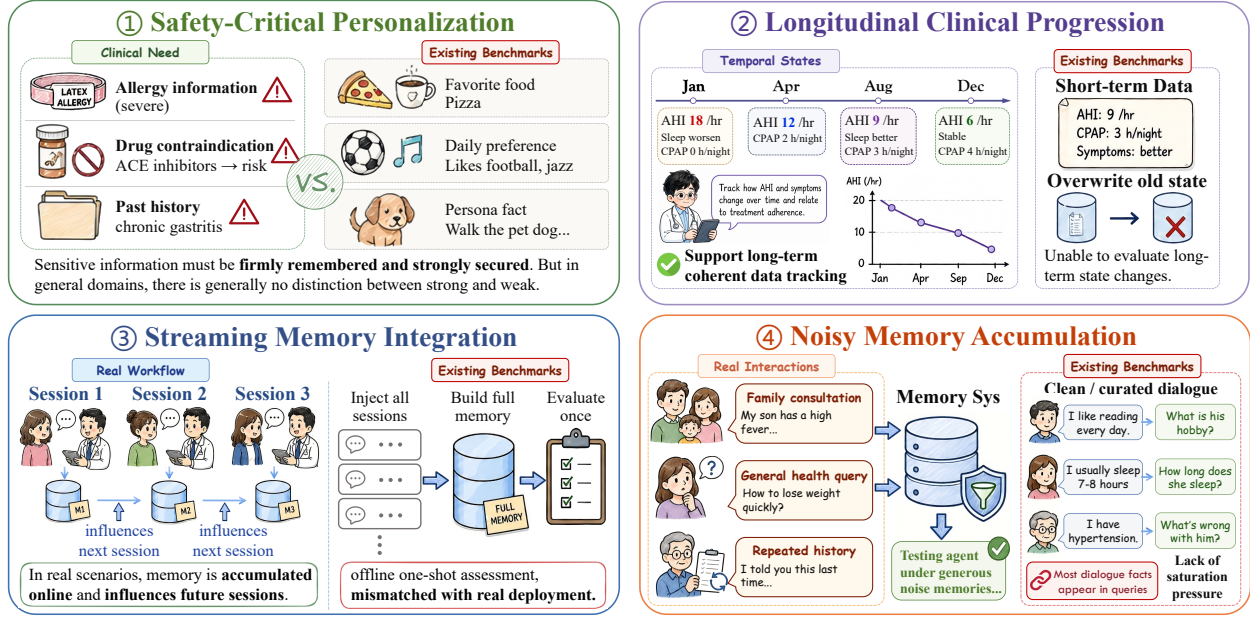


Figure 1: Four representative reasons illustrating why personalized healthcare places stricter demands on agent memory.

these benchmarks have played important roles within their respective scopes, their limitations become increasingly apparent when confronted with realistic, large-scale personalized healthcare scenarios.

As depicted in Figure 1, benchmarking agent memory in personalized healthcare must address four distinctive challenges: ① **Safety-Critical Personalization**: Unlike general conversational facts that carry equal weight, clinical data is highly risk-heterogeneous. Life-threatening details (e.g., severe allergies, medication contraindications) demand strict prioritization and zero-fault retention. ② **Longitudinal Clinical Progression**: Disease management spans months or years. While existing benchmarks merely overwrite short-term states, medical agents must coherently track temporal changes and reason over evolving clinical trajectories. ③ **Streaming Memory Integration**: In production, memory accumulates incrementally, with each session immediately shaping the next. Traditional offline, one-shot evaluations on pre-built memories fundamentally fail to capture this continuous, live dynamic. ④ **Noisy Memory Accumulation**: The persistent influx of clinical records and random user queries (e.g., proxy family consultations, repeated histories) inevitably introduces redundancy and noise. Agents must remain robust against this memory saturation, a pressure entirely absent in clean, curated general datasets. Together, these characteristics expose major blind spots in current evaluation paradigms and call for benchmarks grounded in real medical deployment. These issues raise a central question:

How to address these issues, align with real personalized healthcare scenarios, and evaluate agent memory more comprehensively?

To this end, we propose **MedMemoryBench**, a benchmark for evaluating agent memory in personalized healthcare. To reflect the longitudinal and evolving nature of healthcare, we build a data production pipeline that constructs patient profiles from real clinical portraits, generates a full one-year diagnostic trajectory for each patient, and organizes the process as a directed diagnostic event graph. Patient agents then interact with physician agents grounded in these events, while memory entries are continuously recorded and fed back to preserve dialogue coherence. To capture the retention demands of safety-critical personalization, we introduce *trap events*, such as allergy histories, medication contraindications, and economic constraints, that are embedded into the diagnostic workflow as high-priority signals. We further incorporate a noise augmentation module that injects two types of distracting information, enabling robustness evaluation under prolonged memory accumulation and potential saturation. The entire workflow follows a human-agent collaborative paradigm, with all samples validated, annotated, and refined by medical professionals.

Based on this dataset, we construct approximately 2,000 queries spanning four core dimensions: precise factual retrieval, clinical state updating, implicit clinical reasoning, and system robustness. We conduct a comprehensive evaluation of existing memory methods. Rather than separating memory construction from assessment completely,

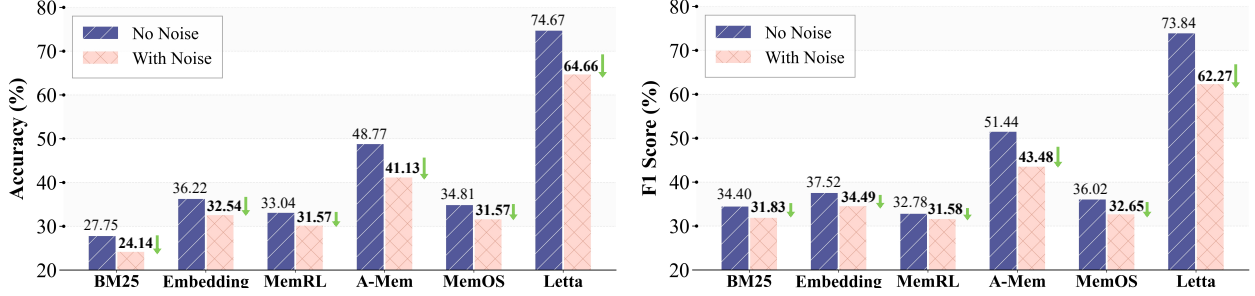


Figure 2: Validation results on LoCoMo under the Efficient and Mixed settings.

MedMemoryBench adopts an *evaluate-while-constructing* protocol that triggers checkpoints as memory accumulates and begins to affect subsequent interactions. Experimental results show that current methods still face substantial limitations: all methods perform poorly on complex medical reasoning tasks that require combining multiple memories, and retrieval remains the key bottleneck limiting performance improvements. More importantly, with the growth of memory and the increase of noise information, the performance of various methods decreases significantly, and long-term memory without adequate filtering often harms overall performance.

Our contributions are as follows:

- We introduce **MedMemoryBench**, a benchmark for evaluating agent memory in personalized healthcare, built through a four-stage human-agent collaborative pipeline with expert medical verification.
- We move beyond static evaluation by proposing a streaming *evaluate-while-constructing* protocol that assesses memory as it accumulates over time, and we provide the first study of *memory saturation* in healthcare agent memory.
- We benchmark representative memory methods and provide a comprehensive analysis, exposing key limitations of current approaches and establishing strong baselines for future research on reliable medical agents.

2 Related Work

Memory Methods and Medical Dialogue Agents. Recent work on agent memory mainly falls into three lines. Retrieval-based methods build on sparse or dense retrieval [17, 27], with graph-enhanced variants such as GraphRAG [7], HippoRAG [10], HippoRAG v2 [11], and Zep [25] improving relational reasoning. Agent-centric memory mechanisms, including Mem0 [5], MemOS [19], Letta/MemGPT [22], A-Mem [37], and MemoryBank [43], emphasize autonomous memory writing, retrieval, and updating. Other designs, such as MemoRAG [23], Mem1 [45], Mem- α [33], and SimpleMem [20], explore consolidation and hierarchical compression. A further line of work targets *memory management and selective forgetting*, aiming to maintain compact, high-quality memory stores through utility-based pruning [9], executive consolidation [24], and experience-driven refinement [3]. In parallel, medical dialogue agents such as Healthcare Agent [26], LD-Agent [18], and AMDS [36] demonstrate the promise of LLMs in healthcare, but they generally treat memory as a static component rather than a long-horizon, continuously updated system.

Agent Memory Benchmarks. Existing benchmarks are largely either *dialogue-centric* or *long-context*. Dialogue-centric benchmarks, such as LoCoMo [21], LongMemEval [34], MemoryAgentBench [14], HaluMem [4], and AMA-Bench [42], evaluate conversational memory, memory operations, or agent trajectories under general-domain settings. Related benchmarks including PersonaMem [15], MemBench [30], MemoryArena [12], Memora [31], and RealTalk [16] further study dynamic profiling, continual memory, and personalized agents. By contrast, long-context or interactive environments such as RULER [13], LongBench [2], WebArena [44], and ALFWorld [28] focus on static context processing or non-medical task environments. Overall, existing benchmarks do not target personalized medical dialogue, where clinical specificity, heterogeneous memory priority, and streaming evaluation are central.

3 Preliminary Study

In our real personalized healthcare deployments, patient interactions are often complex and diverse, giving rise to multiple noteworthy issues for memory mechanisms. Missed retrieval and over-retrieval occur frequently, and demands such as diagnostic reasoning and longitudinal indicator tracking remain difficult to satisfy. More critically, as noise and memory entries accumulate, we observe an instability. Ideally, an agent should function like a human physician by progressively refining its knowledge and evolving its capabilities as clinical experience and the memory store

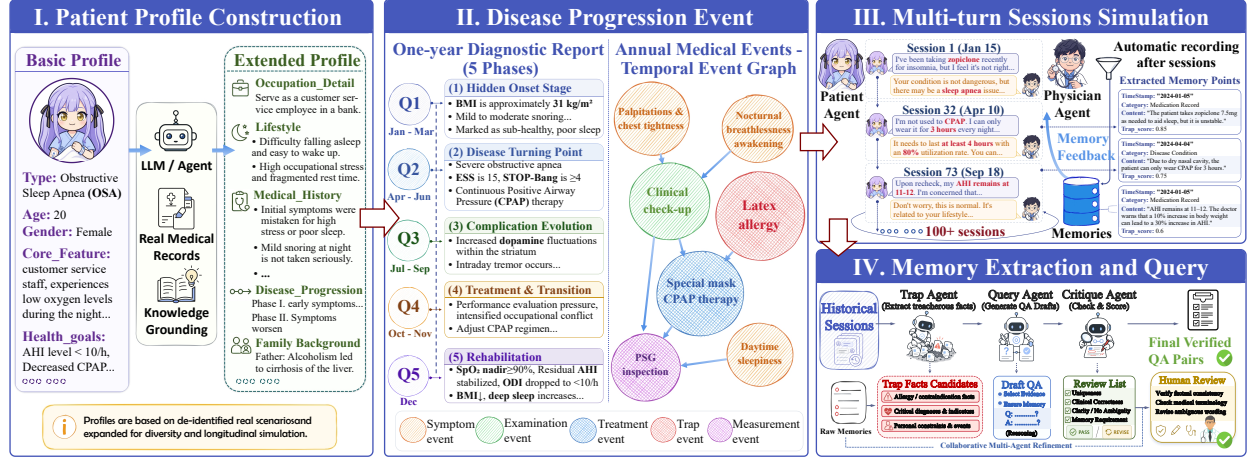


Figure 3: Overview of the MedMemoryBench data construction pipeline. The pipeline consists of patient profile construction, disease progression event generation, multi-turn session simulation, and memory extraction with query construction, with expert verification incorporated throughout.

expand. However, under existing memory architectures, as the number of interaction turns increases, a large volume of semantically similar memory entries are retained. The proliferation of such redundant information causes the retrieval space to be flooded with highly similar noise, making the system prone to extracting redundant or even outdated fragments. We refer to this phenomenon, in which more memory leads to worse performance, as *memory saturation*.

To examine its generality, we reproduced this phenomenon on the existing long-conversation benchmark LoCoMo [21], first under the Efficient setting and then under the Mixed setting, in which GPT-5.1 appends randomly generated daily-life topics after each session. As shown in Figure 2, the injected noise consistently triggers memory saturation and degrades performance across baselines, underscoring the need to evaluate memory methods for robustness and generalizability under sustained information influx.

4 Constructing MedMemoryBench

4.1 Data Construction Pipeline

The MedMemoryBench dataset generation workflow is illustrated in Figure 3. To ensure the fidelity, coherence, and medical reliability of the final data, we design a closed-loop generation pipeline comprising four core stages. Medical expert annotation and manual verification are incorporated at each critical stage.

Patient Profile Construction. We first create 20 representative chronic-disease patient profiles from de-identified real-world medical cases, covering common conditions such as diabetes, cardiovascular diseases, and chronic respiratory diseases. Each profile specifies the disease subtype, initial symptoms, and core clinical characteristics. To improve diversity while preserving privacy, peripheral attributes (e.g., family background, occupation, and lifestyle) are expanded with LLM assistance.

Disease Progression Event Generation. Based on these profiles, we generate expert-verified year-long diagnostic trajectory reports that capture clinically plausible disease progression, treatment, and lifestyle factors, including possible misdiagnosis periods, clinical turning points, complication evolution, lifestyle challenges, and rehabilitation plans. We then decompose each trajectory into disease progression events and organize them into a temporally and causally linked event graph. At the initial stage of the graph, we explicitly inject *trap events*, which are high-priority, patient-specific, and clinically sensitive facts such as past medical history, allergy history, contraindications, economic constraints, and personal preferences. These events serve as key evidence for later question answering and memory construction.

Multi-turn Sessions Simulation. We instantiate patient and physician agents to simulate physician–patient interactions. The patient agent follows the temporal event graph to initiate sessions according to current disease progression, while the physician agent conducts medical assessment and responds to patient complaints. After each session, the system

records a summary containing the session topic, time, and event information as accumulated contextual memory, and feeds it into subsequent interactions to maintain long-term coherence.

Memory Extraction and Query Construction. Finally, we construct query–answer pairs with a multi-agent framework. The *Trap Agent* uses Agentic RAG to retrieve clinically sensitive and highly specific facts from session memories, including allergy and disease history, medication records, indicator changes, and symptom evolution. These facts are often patient-specific and may conflict with generic medical priors. The *Query Agent* formulates customized questions around these facts to force reliance on personal memory rather than prior knowledge; and the *Critique Agent* verifies query uniqueness, memory traceability, and medical professionalism. Medical annotators then perform final verification of trajectories, dialogues, and query–answer pairs, and revise earlier dialogue content when inconsistencies are found.

Human annotation is applied at two key stages: diagnostic trajectory construction and final verification of dialogues and QA pairs. In both stages, annotators check clinical plausibility, correct medical terminology and dosage-related errors, supplement missing critical clinical content when necessary, and ensure answer correctness and standardized medical expression. Detailed annotation rules are provided in [Appendix A.2](#).

4.2 Noise Augmentation

To study how memory saturation affects downstream tasks under realistic redundancy, we construct two interaction modes. In **Efficient Mode**, patients interact with the physician agent only about their own disease status and health information. In **Mixed Mode**, we inject additional auxiliary interactions to simulate the redundant and heterogeneous information commonly observed in real-world healthcare services.

The injected noise takes two forms:

1. **General health knowledge consultations:** queries about routine medical topics, such as when an injection is needed for a cold or how to prevent seasonal influenza. These queries cover 38 preset topic categories, including symptom checks, common medical knowledge, preventive care, and lifestyle advice.
2. **Proxy consultations for family members:** queries raised on behalf of relatives, such as parents, spouses, or children. Each patient is associated with 2–4 virtual family members who have independent health profiles, including disease history, medication records, and recent health changes, and the patient asks questions in a proxy form (e.g., “my father recently has high blood pressure...”).

For each patient, both types of noise sessions are randomly inserted into the original chronological dialogue sequence, and their extracted memory summaries are marked with an `is_noise:true` flag.

4.3 Statistics

Following the above pipeline, MedMemoryBench contains 20 chronic-disease patients, each with approximately 100 physician–patient sessions of 6–10 turns and corresponding memory summaries. The dataset totals about 20.74 million tokens and 16,000 dialogue turns, and includes nearly 100 queries per patient, for 1,939 queries overall. The Mixed setting further appends roughly 200 additional sessions per patient, increasing the total number of dialogue turns to approximately 42,000.

5 Evaluation Framework

5.1 Query Design

We define six query types spanning precise factual retrieval, clinical state updating, and implicit clinical reasoning. They are grouped into three capability bands: *precise memory extraction* (EEM, TLA), *temporal conflict resolution* (SUA), and *medical knowledge reasoning* (MQ, IG, MCD). [Table 1](#) summarizes their evaluation targets, answer formats, and task definitions. The LLM-as-Judge prompts are provided in [Appendix D.1](#).

Among them, IG and MCD place the strongest demands on patient-specific reasoning. IG requires clinically grounded responses that preserve critical personalized constraints. MCD is evaluated by three LLM-as-Judge criteria: **Node Coverage Rate (NCR)**, which checks whether the required reasoning nodes are covered; **Causal Relation Correctness (CRC)**, which checks whether the causal or temporal links among nodes are correct; and **Chain Completeness (CC)**, which checks whether the overall reasoning chain is complete. A response is counted as correct only if all three criteria are satisfied. Accordingly, all query types are evaluated using binary scoring, where each response receives a score of 1 if correct and 0 otherwise. Robustness is assessed by comparing performance under the Efficient and Mixed settings in the streaming protocol.

Table 1: Definition of query types in MedMemoryBench. **Tags:** **S** = String Match, **L** = LLM-as-Judge, and **O** = Option Match.

Query Type	Evaluation	Answer Format	Description
Entity Exact Match (EEM)	S	Entity or value	Answer factual questions involving specific entities.
Temporal Location Accuracy (TLA)	S L	Time point, time span, or event	Identify when a clinical event occurred, or determine what happened at a given time.
State Update Accuracy (SUA)	L	Latest state or state trajectory	Return the latest patient state or describe the temporal evolution of clinical attributes.
Multiple Choice (MQ)	O	Discrete option	Select the correct option using medical knowledge and patient-specific constraints.
Inference Generation (IG)	L	Open-ended text	Generate clinically grounded responses using patient-specific constraints and trap information.
Multi-hop Clinical Deduction (MCD)	L	Open-ended reasoning chain	Construct a complete causal reasoning chain across 3–5 memory nodes.

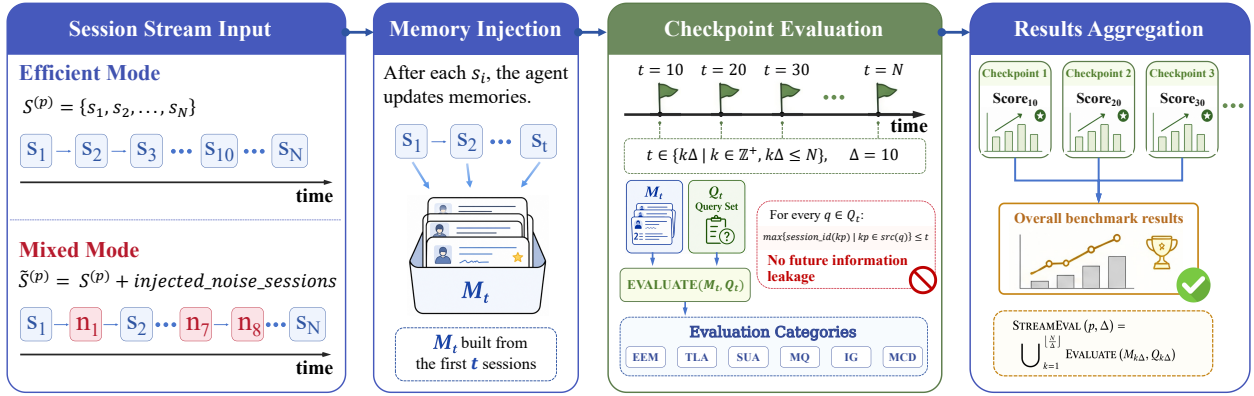


Figure 4: Overview of the MedMemoryBench evaluation framework. The figure summarizes the streaming *evaluate-while-constructing* protocol used to assess memory methods under the Efficient and Mixed settings.

5.2 Evaluation Protocol

Unlike static evaluation that first constructs all memory and then evaluates, MedMemoryBench adopts an *evaluate-while-constructing* protocol that better reflects real deployment (Figure 4). Under this protocol, a memory method receives sessions strictly in chronological order and must answer checkpoint queries using only the memory accumulated so far. For patient p , let the complete session sequence be $\mathcal{S}^{(p)} = \{s_1, s_2, \dots, s_N\}$. We evaluate every Δ sessions, with $\Delta = 10$, at checkpoints $t \in \{k\Delta \mid k \in \mathbb{Z}^+, k\Delta \leq N\}$. For patients in the Mixed setting, randomly injected auxiliary sessions $\tilde{s}$ yield an extended sequence $\tilde{\mathcal{S}}^{(p)}$.

At checkpoint t , let $\mathcal{M}_t$ denote the memory state built from the first t sessions and $\mathcal{Q}_t$ the query set assigned to that checkpoint. The streaming protocol is defined as:

$$\text{STREAMEVAL}(p, \Delta) = \bigcup_{k=1}^{\lfloor N/\Delta \rfloor} \text{EVALUATE}(\mathcal{M}_{k\Delta}, \mathcal{Q}_{k\Delta}) \quad (1)$$

where $\text{EVALUATE}(\mathcal{M}, \mathcal{Q})$ returns the evaluation results of query set $\mathcal{Q}$ under memory state $\mathcal{M}$. The agent then continues updating memory until the next checkpoint.

Each query $q \in \mathcal{Q}_t$ is constrained so that all of its source knowledge points $\text{src}(q)$ appear no later than checkpoint t :

$$\forall q \in \mathcal{Q}_t, \quad \max\{\text{session_id}(kp) \mid kp \in \text{src}(q)\} \leq t \quad (2)$$

This prevents future information leakage by ensuring that all answer-relevant knowledge has already entered memory at evaluation time.

Table 2: Main results on MedMemoryBench under the Efficient and Mixed settings. The metric is response accuracy across six query types, with higher values indicating better performance. Best and second-best results are marked in **bold** and underlined, respectively.

Methods	EEM		TLA		SUA		MQ		IG		MCD		Avg.	
	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.
Unstructured														
Long-Context	50.64	38.15	40.26	<u>38.22</u>	76.62	65.75	73.86	54.55	<u>43.60</u>	25.60	24.50	10.25	51.58	38.75
BM25	36.70	35.50	12.70	10.50	45.50	43.00	73.30	60.80	21.16	18.00	18.85	5.38	34.70	28.86
Embedding	51.75	55.25	20.75	18.96	55.00	<u>65.00</u>	71.86	65.32	28.96	24.00	<u>24.60</u>	16.12	42.15	40.78
Graph-Structured														
GraphRAG	37.50	43.00	23.21	37.50	46.43	44.00	60.00	51.25	23.21	10.06	11.53	6.45	33.65	32.03
HippoRAG-v2	43.50	40.00	18.70	23.00	31.00	38.00	69.08	57.58	25.44	10.00	14.25	5.20	33.69	28.96
ReMem	32.50	30.00	18.00	18.50	56.00	62.00	57.79	55.78	23.80	21.00	15.50	9.37	33.93	<u>32.78</u>
Flat-Extracted														
Mem0	44.00	43.50	20.50	22.50	53.50	60.00	64.82	63.31	30.98	17.50	21.40	8.14	39.20	35.83
A-Mem	46.00	20.50	27.00	26.50	52.00	28.00	71.61	64.32	40.55	25.50	31.93	10.80	44.84	29.27
MemOS	<u>58.50</u>	49.00	43.00	40.50	66.00	58.00	71.10	<u>67.83</u>	35.11	26.25	15.23	4.30	48.16	<u>40.98</u>
MemRL	55.00	40.00	43.00	33.50	55.00	50.00	70.10	<u>67.83</u>	43.07	<u>32.00</u>	12.04	<u>13.97</u>	46.37	<u>37.45</u>
LightMem	49.25	24.00	35.75	23.50	58.00	37.00	67.10	59.29	18.60	14.00	13.25	6.07	40.33	27.31
Multi-Store														
Letta	58.95	40.50	<u>40.74</u>	23.00	<u>68.52</u>	52.00	66.97	70.35	52.87	49.50	19.20	<u>13.97</u>	<u>51.21</u>	41.55
MIRIX	49.75	42.25	25.25	10.78	<u>44.50</u>	46.00	<u>73.37</u>	54.75	31.49	22.33	16.37	8.10	40.12	30.70

6 Experiment

6.1 Experimental Setup

Models. All baselines employ GPT-5.1 as the backbone model, with Claude-4-Sonnet serving as the judge model. The embedding model is uniformly set to BGE-small-v1.5. We report experimental results with additional backbone and judge models in [Appendix C](#).

Baselines. We evaluate mainstream open-source memory methods, including Mem0 [5], A-Mem [37], MemOS [19], Letta [22], MIRIX [32], MemRL [41], ReMem [29], LightMem [8], and HippoRAG-v2 [11], plus three RAG baselines: BM25 RAG [27], Embedding RAG [17], and GraphRAG [7]. We also include a Long-Context control that directly feeds the raw dialogue history, whereas all other memory baselines see only their stored memory entries. Methods are grouped into Unstructured, Graph-Structured, Flat-Extracted, and Multi-Store. The default number of retrieved passages is 5, with a chunk size of 4,096.

6.2 Overall Performance and Main Findings

[Table 2](#) summarizes overall performance on MedMemoryBench. Existing methods remain far from satisfactory across query types, with the largest weaknesses appearing on reasoning-intensive tasks such as IG and MCD. We highlight three main findings:

Moderate memory granularity works best. Under the Efficient setting, methods with medium-grained memories, such as A-Mem, MemOS, and Letta, perform best overall. By contrast, Mem0 fragments information into many small entries and weakens contextual links, whereas MIRIX packs too much content into each entry and reduces retrieval precision. Graph-structured methods also lag behind, suggesting that current general-purpose graph construction pipelines are not well adapted to medical memories with high overlap, strong temporal dependencies, and fine-grained attribute updates. Therefore, memory design must balance local contextual completeness with the locatability of critical facts, aligning retrieval granularity with the information density of memory entries.

Retrieval is the main bottleneck. By inspecting the memory construction and retrieval details of different methods, we find that retrieval failure is a key factor behind their performance gaps. Letta’s advantage is largely attributable to its Core Memory, which keeps critical information persistently visible in the system prompt and therefore reduces reliance

Table 3: Response quality and safety metrics. Higher Recall, F1 score, and ROUGE-L indicate better coverage and faithfulness, while lower Hallucination indicates safer behavior. Hippo-v2 refers to HippoRAG-v2. Best and second-best results are marked in **bold** and underlined, respectively.

Metric	Unstructured		Graph-Structured			Flat-Extracted					Multi-Store	
	BM25	Embedding	GraphRAG	Hippo-v2	ReMem	Mem0	A-Mem	MemOS	MemRL	LightMem	Letta	MIRIX
Recall (R@k)↑	0.3563	0.5043	0.2281	0.3561	0.3641	0.3401	0.2847	<u>0.5259</u>	0.5451	0.3784	0.2812	0.4727
F1 score↑	0.2021	0.2167	0.2287	0.2800	0.1890	0.1358	0.2265	0.2616	0.2761	<u>0.2771</u>	0.2624	0.2053
ROUGE-L↑	0.2149	0.2246	0.2332	0.2862	0.1907	0.2072	0.2416	0.2625	<u>0.2717</u>	0.2715	0.2577	0.2107
Hallucination↓	0.7181	0.7288	0.7947	0.9209	0.5993	0.6869	0.7719	0.7974	0.8314	0.8050	0.8001	<u>0.6775</u>

on precise retrieval; its agent-driven memory management further improves access quality. In contrast, many other methods build richer memory stores but still fail to reliably retrieve the most relevant evidence from large memory pools, especially for IG and MCD.

Memory saturation degrades reasoning robustness. Under the Mixed setting, all methods show performance degradation, while the magnitude of decline varies across methods. Most methods suffer substantial losses on IG and MCD. The underlying reason is that memory scenarios are fundamentally different from RAG scenarios. Unlike the heterogeneous and diverse data in RAG, data in memory scenarios gradually accumulates and becomes more homogeneous. On the one hand, redundant memories impose stricter demands on retrieval precision; on the other hand, they also disrupt memory construction itself. For methods that rely on selective memory updates or conflict resolution, noisy sessions can distort judgments about what should be retained, causing key information to be missed during memory building. This result highlights the fragility of current memory methods in the real environment of sustained information growth.

6.3 Response Quality and Hallucination Analysis

Table 3 further evaluates response quality and safety from four perspectives. Recall (R@k) measures how well the retrieved memory set covers the true memories required by the query, reflecting memory retrieval completeness. F1 score evaluates the token-level overlap between generated answers and reference responses, capturing overall answer correctness. ROUGE-L measures longest common subsequence overlap and is used to assess response faithfulness at the sequence level. In medical scenarios, a memory system must not only answer correctly, but also know when to abstain under insufficient evidence. Therefore, hallucination measures, among all incorrect responses, the proportion in which the model does not abstain but instead gives a confident wrong answer, directly reflecting safety risk in medical deployment.

Overall, current methods show a noticeable gap between evidence coverage and final answer reliability, indicating that errors arise not only from missing memories but also from imperfect grounding and answer composition after retrieval. Moreover, higher Recall does not necessarily translate into better response quality, and hallucination is not fully determined by retrieval or generation quality alone. This is because memory systems are not single-step retrieval modules, but pipelines involving memory construction, evidence filtering, answer composition, and reasoning, where errors can accumulate across stages. The generally high hallucination rates further suggest that once memory support is insufficient or ambiguous, models tend to overcommit rather than refuse, which is particularly problematic in healthcare settings.

6.4 Ablation Study

We conduct ablations on retrieval top- k and chunk size for memory methods on MedMemoryBench under the Efficient setting. As shown in Figure 5, increasing top- k generally improves performance by exposing the model to more candidate memories, although the gains vary across methods. A-Mem benefits markedly from larger top- k values because its memory entries are relatively structured and information rich, allowing the model to exploit additional retrieved evidence without losing the central clinical thread. By contrast, Letta depends more on its Core Memory mechanism, which keeps highly salient information persistently available and therefore reduces the marginal utility of retrieving more external entries. LightMem improves from top-2 to top-10 but declines at top-20, suggesting that excessive evidence can dilute key facts and cause the model to attend to plausible but irrelevant memories.

The chunk size ablation reveals a trade-off between contextual completeness and noise control. Larger chunks generally provide more complete medical context and improve performance, but the trend is not monotonic: extra context can also dilute critical information, especially for methods with already dense memory representations.

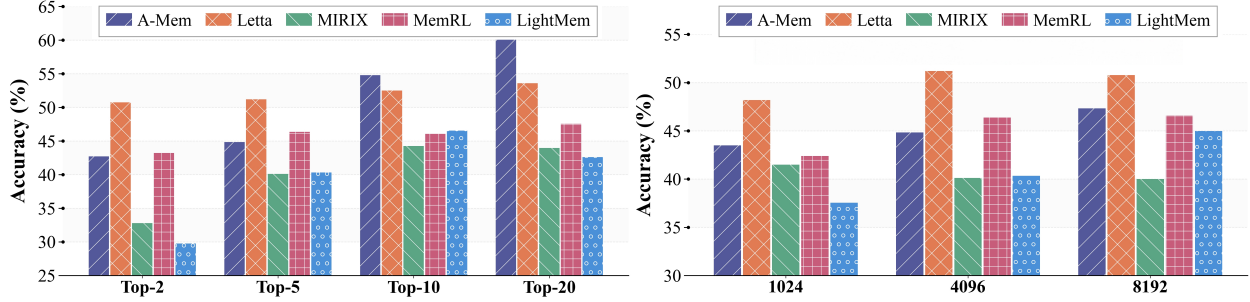


Figure 5: Performance on MedMemoryBench with different retrieval $\text{top-}k$ values and **chunk sizes**.

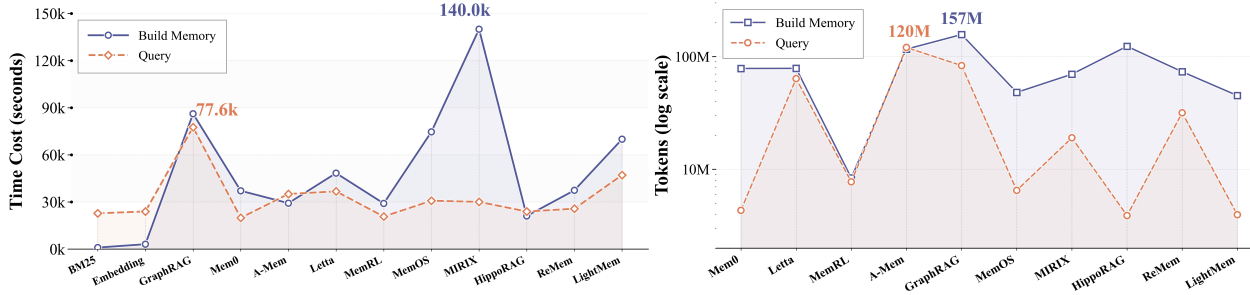


Figure 6: **Time and token costs** for memory building and query answering across representative memory methods. The values in the figure indicate the highest point.

6.5 Efficiency and Cost Analysis

We evaluate cost across two stages, namely memory building and query answering. For token consumption, we count only LLM API tokens used for text processing, excluding embedding models. Figure 6 shows that current memory methods still leave substantial room for improvement in cost efficiency.

For time cost, variance is much larger in the building stage than in the query stage. Some methods require additional structural processing when writing and organizing memories, including entity extraction, relation construction, conflict checking, summarization, or memory consolidation, raising build cost to more than five times that of RAG baselines. Query-time costs, in contrast, are relatively similar across methods because most systems eventually reduce the problem to retrieving a limited set of memories and generating a response. This gap suggests that long-term memory organization and maintenance is the main efficiency bottleneck.

The same pattern appears in token consumption: memory building dominates cost for most methods, with near order-of-magnitude differences across approaches. MemRL achieves a favorable balance by computing reward values directly and reducing model calls, whereas graph-structured methods incur substantial token overhead from graph construction, edge rebuilding, and node extraction. Combined with the main results, these findings show that several high-cost methods do not deliver commensurate gains, revealing a clear efficiency–performance imbalance in current approaches.

7 Conclusion

We present **MedMemoryBench**, a benchmark for evaluating agent memory in personalized healthcare. Through a systematic data construction pipeline, streaming evaluation protocol, and noise augmentation module, our benchmark addresses critical gaps in existing benchmarks and reveals key challenges faced by current memory methods: memory granularity, retrieval effectiveness, and robustness under information redundancy. We hope MedMemoryBench provides valuable guidance for future agent memory research toward achieving robust and reliable long-term memory augmentation.¹

¹Limitation and future work are discussed in [Appendix G](#).

References

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Table 4: Detailed statistics of MedMemoryBench.

Original Dialogue Statistics		QA Benchmark Statistics				Noise Statistics	
Total sessions	2,020	EEM	400	MQ	398	Health-topic categories	98
Total dialogue turns	15,988	TLA	374	IG	381	Avg. sessions per topic	20.41
Avg. turns per session	7.915	SUA	195	MCD	191	Family members	5
Avg. tokens per session	10,266.37	Total			1,939	Sessions per family member	20
Avg. memory items per session	2.47	Avg. memory items / query			2.62		
Avg. tokens per memory item	54.15						

A Details of MedMemoryBench

A.1 Detailed Statistical Description

As shown in Table 4, we provide more detailed statistics for MedMemoryBench, covering the original dialogue corpus, the QA benchmark composition, and the two auxiliary noise sources used during data construction.

A.2 Medical Annotation Details

This subsection summarizes the human annotation process used in our data production pipeline.

Annotation scope. In the diagnosis trajectory reporting stage, annotators were responsible for three tasks: (1) verifying the medical plausibility of disease progression to ensure that the trajectory followed clinically reasonable patterns; (2) correcting professional terminology and medication dosages in diagnosis and treatment plans to remove medical expression errors; and (3) supplementing potentially missing clinical key events, such as necessary examinations, typical complications, or other essential decision points.

In the later verification stage, including both raw dialogue review and QA-pair post-processing, annotators were responsible for three additional tasks: (1) verifying answer correctness so that each reference answer was consistent with the dialogue-derived memory content; (2) reviewing the accuracy and standardization of medical expressions, including drug names, dosage units, laboratory indicators, and answer formatting; and (3) marking ambiguous or overly simple queries and, when necessary, directly revising the corresponding questions and answers.

Annotators. All annotators involved in constructing MedMemoryBench were engineering staff from the medical division of Ant Afu. Their personal information is confidential and therefore not disclosed.

A.3 Human-Agent Consistency

For the query types evaluated with LLM-as-Judge, we randomly sampled 200 query-response pairs from the experimental outputs of three representative baselines: MemOS, A-Mem, and Letta. Human annotators then independently assigned a binary judgment (correct or incorrect) to each model response based on the reference answer, and we compared these annotations against the decisions produced by the LLM judge to assess human-agent consistency.

We report five metrics for this comparison: Agreement Rate, which measures the proportion of directly matched judgments; Cohen’s Kappa (κ), which adjusts inter-rater agreement for chance effects [6]; and Precision, Recall, and F1 score, computed by treating human annotation as the reference standard.

As shown in Figure 7 and Table 5, the overall consistency is high, indicating that the LLM-as-Judge scores are broadly aligned with human judgment. The IG query type exhibits relatively weaker consistency, most notably with a Recall of only 0.58. Manual inspection shows that many disagreement cases arise when a model response provides a clinically plausible inference without explicitly grounding the answer in patient-specific memory details. Human annotators are more likely to accept such responses, whereas the LLM judge applies a stricter standard for memory grounding.

B Supplementary Instructions for Experimental Setup

B.1 LoCoMo

In Section 3, we use the widely adopted long-dialogue benchmark LoCoMo to verify the memory saturation phenomenon. Below, we provide the detailed experimental configuration.

Table 5: Human-agent consistency results across query types evaluated with LLM-as-Judge. Higher Agreement, κ , Precision, Recall, and F1 score indicate stronger alignment with human judgment.

Query Type	#Samples	Agreement (%)	Cohen’s κ	Precision	Recall	F1 score
TLA	50	86.0	0.69	0.82	0.78	0.80
SUA	30	90.0	0.80	0.87	0.93	0.90
IG	60	80.0	0.559	0.88	0.58	0.70
MCD	60	95.0	0.81	1.00	0.73	0.84
Overall	200	87.5	0.72	0.88	0.73	0.80

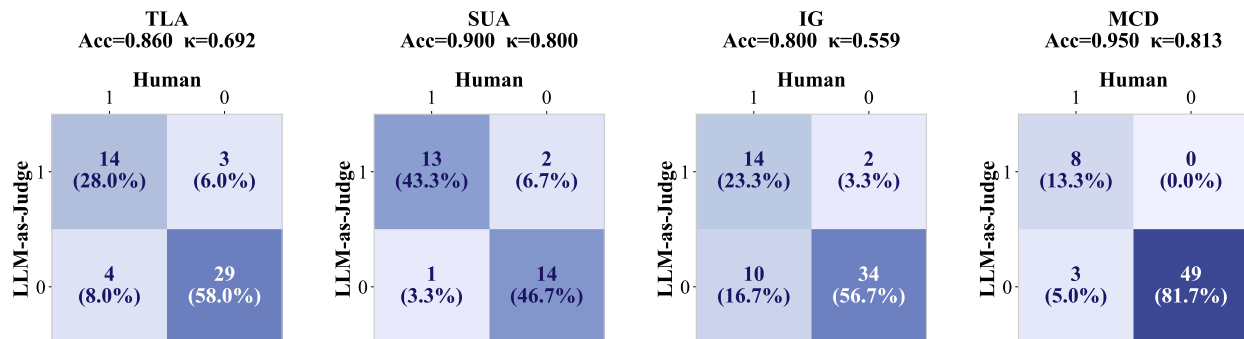


Figure 7: Human-agent consistency analysis based on confusion matrices across query types evaluated with LLM-as-Judge.

Evaluation protocol. We follow the evaluation protocol of the original LoCoMo paper. The evaluation covers all five query categories: single-hop factual retrieval (single-hop, category 4), multi-hop information integration (multi-hop, category 1), temporal reasoning (temporal, category 2), open-domain reasoning (open-domain, category 3), and adversarial questions (adversarial, category 5).

Prompt adjustment. The official LoCoMo evaluation uses simple open-ended prompts without explicit output constraints. In practice, this setting encourages LLMs to produce unnecessarily long responses. We therefore add a concise-answer instruction for each query type so that the model is discouraged from generating lengthy explanations.

Metric. Consistent with the official evaluation, we use F1 score to measure the overlap between model responses and reference answers. A response is counted as correct when $F1 \geq 0.5$, and the final result is reported as the average accuracy over all questions.

B.2 MedMemoryBench

For all evaluations on MedMemoryBench, we use the OpenAI API. For all models, the temperature is set to 0.3 and the maximum number of completion tokens is set to 10,240. For the long-context baseline, which takes the raw session content directly as input, we set the context window to 128k. All experiments were conducted on NVIDIA H200 servers.

C Additional Experimental Results

C.1 Detailed Results on More Metrics

In addition to the main results reported in the paper, we provide a more fine-grained breakdown of four supplementary metrics: Recall, F1 score, ROUGE-L, and Hallucination. For each metric, we report the performance of all methods on every query type in MedMemoryBench, so that differences in retrieval quality, answer overlap, and unsupported generation can be examined in greater detail. The detailed results are shown in Table 6 and Table 7.

Table 6: Detailed results on Recall and F1 score across query types in MedMemoryBench. Best and second-best results are marked in **bold** and underlined, respectively.

Methods	EEM		TLA		SUA		MQ	IG		MCD	
	Recall↑	F1↑	Recall↑	F1↑	Recall↑	F1↑	Recall↑	Recall↑	F1↑	Recall↑	F1↑
Unstructured											
BM25-RAG	51.50	46.50	45.00	23.32	31.49	6.36	40.83	25.67	10.64	19.30	14.23
Embedding-RAG	69.50	58.79	<u>69.20</u>	19.65	<u>36.95</u>	6.70	47.95	<u>51.50</u>	10.26	27.49	12.97
Graph-Structured											
GraphRAG	53.57	49.97	32.14	22.91	13.87	7.23	7.73	11.61	20.86	17.95	13.36
HippoRAG-v2	44.00	55.68	44.00	22.42	26.42	12.28	35.39	40.50	22.88	23.33	26.76
ReMem	46.50	34.03	54.00	19.61	21.30	10.39	33.75	36.83	20.11	26.11	10.38
Flat-Extracted											
Mem0	58.50	10.38	63.00	13.00	18.11	<u>10.73</u>	23.12	23.42	20.75	17.92	13.05
A-Mem	36.00	56.11	38.00	28.80	19.02	5.17	24.20	28.58	10.05	25.00	13.13
MemOS	<u>77.00</u>	<u>59.56</u>	75.00	32.17	30.90	7.63	42.21	50.12	18.76	40.32	12.67
MemRL	80.50	61.93	66.00	<u>33.90</u>	35.12	6.73	<u>45.23</u>	52.71	20.22	47.49	15.26
LightMem	63.50	57.42	68.50	37.45	18.58	8.81	26.13	28.58	20.71	21.77	14.16
Multi-Compartment											
Letta	41.50	54.28	38.50	29.08	20.14	10.64	18.46	23.96	<u>21.04</u>	26.16	<u>16.18</u>
MIRIX	56.50	49.31	52.00	19.07	42.53	5.93	43.89	41.50	16.02	<u>47.22</u>	12.34

Overall, the supplementary metrics reveal a clear trade-off between retrieval coverage, answer faithfulness, and response conservativeness. Flat-extracted methods remain strong on relatively direct retrieval settings such as EEM and TLA, where MemRL, MemOS, and LightMem achieve the best Recall/F1 or ROUGE-L values, while graph-based methods become more competitive on structurally harder settings such as SUA, IG, and especially MCD, with HippoRAG-v2 obtaining the best F1 or ROUGE-L in several categories. At the same time, these gains do not always translate into safer behavior: ReMem consistently yields the lowest hallucination rates, but its answer-overlap metrics are usually lower, indicating that conservative generation can reduce unsupported outputs at the cost of coverage and final-answer completeness.

Recall. Recall measures whether the memory system can retrieve the key information required to answer a query from its constructed memory store. We evaluate this metric using an LLM-as-Judge protocol. Specifically, for each query q_i , the dataset provides a set of ground-truth key memory snippets required to answer the query, denoted by $\mathcal{K}_i = \{k_1, k_2, \dots, k_n\}$ (i.e., `source_key_points`). During evaluation, the retrieved memory content returned by the system, denoted by R_i , is compared against each key snippet k_j by the LLM judge, which determines whether the core semantics of k_j are covered by R_i . Semantic equivalence is sufficient; exact surface-form matching is not required. The recall for an individual query is defined as:

$$\text{Recall}(q_i) = \frac{|\{k_j \in \mathcal{K}_i \mid \text{covered}(k_j, R_i) = \text{True}\}|}{|\mathcal{K}_i|}$$

The final score is reported by averaging over queries within each query type, as well as over the full benchmark. This metric directly reflects the completeness of memory retrieval, independently of downstream answer generation quality.

F1 Score. F1 score measures the token-level overlap between a model-generated answer and the reference answer. Given a generated answer $\hat{y}$ and a reference answer y , we tokenize both answers and remove punctuation-only tokens, obtaining token sequences $\hat{T}$ and T , respectively. Precision, recall, and F1 are then computed based on multiset overlap:

$$P = \frac{|\hat{T} \cap T|}{|\hat{T}|}, \quad R = \frac{|\hat{T} \cap T|}{|T|}, \quad F_1 = \frac{2PR}{P + R}$$

where $|\hat{T} \cap T|$ denotes the number of shared tokens between $\hat{T}$ and T , with each token counted by the minimum of its occurrence frequencies in the two multisets.

ROUGE-L. ROUGE-L measures sequence-level similarity between the generated answer and the reference answer based on the *longest common subsequence* (LCS). Unlike F1 score, which focuses on bag-of-words overlap, ROUGE-L

Table 7: Detailed results on ROUGE-L and Hallucination across query types in MedMemoryBench. Best and second-best results are marked in **bold** and underlined, respectively.

Methods	EEM		TLA		SUA		IG		MCD	
	ROUGE-L $\uparrow$	Hall. $\downarrow$	ROUGE-L $\uparrow$	Hall. $\downarrow$	ROUGE-L $\uparrow$	Hall. $\downarrow$	ROUGE-L $\uparrow$	Hall. $\downarrow$	ROUGE-L $\uparrow$	Hall. $\downarrow$
Unstructured										
BM25-RAG	54.85	88.10	33.75	74.71	6.51	82.57	5.46	54.95	6.89	58.71
Embedding-RAG	66.64	96.89	27.41	75.71	6.48	92.13	5.33	51.77	6.42	47.92
Graph-Structured										
GraphRAG	53.70	94.29	33.85	86.05	8.58	80.00	<u>12.91</u>	67.44	7.54	69.57
HippoRAG-v2	64.70	96.90	31.23	84.45	14.85	91.30	16.13	93.58	16.21	94.24
ReMem	40.29	74.07	25.89	<u>58.54</u>	11.36	65.91	11.98	65.58	5.82	35.56
Flat-Extracted										
Mem0	57.16	82.30	17.62	54.30	10.24	84.85	11.73	75.76	6.84	46.24
A-Mem	64.95	97.69	39.03	88.70	5.02	85.42	5.23	<u>53.39</u>	6.59	60.77
MemOS	<u>66.82</u>	93.37	40.36	81.58	7.42	85.29	10.27	<u>78.22</u>	6.40	60.22
MemRL	68.24	95.00	<u>41.93</u>	80.26	7.09	88.89	11.12	92.04	7.48	59.52
LightMem	63.90	88.67	43.94	82.10	8.78	89.13	11.98	80.47	7.15	62.11
Multi-Compartment										
Letta	59.38	80.45	36.37	73.44	<u>11.37</u>	88.24	12.52	84.13	<u>9.20</u>	73.77
MIRIX	57.96	<u>76.62</u>	26.72	65.19	5.96	<u>78.38</u>	8.53	76.84	6.20	<u>41.71</u>

preserves token order information. We use character-level ROUGE-L F-measure. Let X be the character sequence of the generated answer with length m , and let Y be the character sequence of the reference answer with length n . The metric is computed as:

$$P_{\text{lcs}} = \frac{\text{LCS}(X, Y)}{m}, \quad R_{\text{lcs}} = \frac{\text{LCS}(X, Y)}{n}, \quad \text{ROUGE-L} = \frac{2 \cdot P_{\text{lcs}} \cdot R_{\text{lcs}}}{P_{\text{lcs}} + R_{\text{lcs}}}$$

Hallucination Rate. Hallucination rate measures, among incorrect responses, the proportion of cases in which the model produces a confident but unsupported answer rather than appropriately refusing to answer. This metric reflects the reliability of a memory system: when the answer cannot be determined, an ideal system should abstain instead of fabricating information. Formally, we define:

$$\text{Hallucination Rate} = \frac{N_{\text{non-refusal incorrect}}}{N_{\text{total incorrect}}}$$

where $N_{\text{total incorrect}}$ denotes the total number of incorrectly answered queries, and $N_{\text{non-refusal incorrect}}$ denotes the subset of those cases in which the model does not emit a refusal signal. Refusal signals are detected through keyword-pattern matching that covers common abstention expressions, such as indicating that relevant information was not found, that the answer cannot be determined, or that the available information is insufficient.

C.2 Results with Different Backbones

To further evaluate the robustness of the compared methods to the choice of foundation model, we replace the default backbone used by all baselines and rerun the experiments under both the Efficient and Mixed settings. Specifically, we use Qwen3-235B-A22B as the alternative backbone model. The corresponding results are reported in Table 8.

The backbone substitution changes absolute scores but does not overturn the main picture of method competitiveness. In particular, Letta remains the strongest overall system under both settings, while Long-context and Embedding-RAG benefit more visibly from the stronger backbone on several query types, suggesting that generative capacity can partially compensate for weaker memory organization. By contrast, several extracted-memory methods show larger drops in the Mixed setting, which indicates that their effectiveness depends not only on memory design but also on the interaction between retrieval outputs and backbone-specific generation behavior.

C.3 Results with Different Judge Models

To examine whether the default judge model, Claude-sonnet-4, introduces systematic bias in evaluation, we additionally repeat all query types that require LLM-as-Judge with a different judge model. As shown in Table 9, the overall

Table 8: Results on MedMemoryBench with Qwen3-235B-A22B as the backbone model under the Efficient and Mixed settings. The metric is response accuracy across six query types, with higher values indicating better performance. Best and second-best results are marked in **bold** and underlined, respectively.

Methods	EEM		TLA		SUA		MQ		IG		MCD		Avg.	
	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.	Eff.	Mix.
Unstructured														
Long-context	48.20	41.33	<u>42.50</u>	<u>36.78</u>	64.00	64.75	65.86	45.33	37.60	<u>28.00</u>	26.53	14.97	<u>47.45</u>	38.53
BM25-RAG	38.50	37.24	22.00	18.19	48.50	45.84	68.84	57.11	27.70	23.56	16.23	4.63	36.96	31.10
Embedding-RAG	<u>61.25</u>	65.40	32.25	29.47	59.50	70.32	49.74	45.21	24.20	20.05	14.71	9.64	40.28	40.02
Graph-Structured														
GraphRAG	34.00	38.99	26.20	42.33	47.80	45.30	44.00	37.58	27.74	11.95	12.55	7.02	32.05	30.53
HippoRAG-v2	48.85	44.92	15.67	19.27	33.55	41.13	49.33	41.12	20.00	7.86	18.25	6.66	30.94	26.83
ReMem	45.78	42.26	23.20	23.85	38.75	42.90	55.00	53.09	10.80	9.53	12.00	7.25	30.92	29.81
Flat-Extracted														
Mem0	37.24	36.82	22.75	24.97	58.00	<u>65.05</u>	65.24	<u>63.73</u>	30.00	16.95	18.50	7.04	38.67	35.76
A-Mem	52.00	23.88	18.75	18.92	59.50	39.55	<u>66.33</u>	57.82	<u>40.83</u>	14.38	<u>21.46</u>	4.43	43.15	26.50
MemOS	61.50	<u>51.51</u>	29.50	27.79	46.00	40.42	60.70	57.91	10.66	7.97	12.04	3.40	36.73	31.50
MemRL	56.00	40.73	32.50	25.32	42.00	38.18	47.23	45.71	18.78	13.95	14.08	16.34	35.10	30.04
LightMem	49.00	23.18	28.78	18.40	<u>62.00</u>	32.04	65.44	59.58	19.10	25.68	9.67	7.26	39.00	27.69
Multi-Store														
Letta	44.91	30.85	53.30	30.09	60.00	45.53	62.55	65.71	59.78	55.98	14.50	10.55	49.17	<u>39.79</u>
MIRIX	44.78	38.03	34.00	14.51	29.24	30.23	65.25	48.69	33.70	23.90	18.00	8.91	37.50	27.38

Table 9: Results obtained with different judge models. Each value reports the overall score (%) under the corresponding judge model.

Judge Model	Embedding	Mem0	A-Mem	Letta	MemRL	ReMem
claude-sonnet-4	42.15	39.20	44.84	51.21	46.37	33.93
gpt-5.1	40.78	35.03	44.34	47.69	42.98	35.07
gpt-4o	43.30	38.93	45.12	54.13	47.20	35.29
gemini-2.5-pro	46.11	48.32	49.91	57.75	50.02	37.79

conclusions remain largely unchanged, suggesting that the benchmark results are not sensitive to the specific judge model choice.

More specifically, the relative ordering is highly stable across judge models: Letta remains the top-performing baseline under all four judges, while ReMem stays at the bottom, and the middle-tier methods also exhibit only limited rank fluctuation. Although Gemini-2.5-Pro assigns uniformly higher absolute scores than the other judges, this shift is largely monotonic rather than selective, which suggests that changing the judge mainly affects score calibration instead of the comparative conclusions.

C.4 Results by Chronic Disease Category

MedMemoryBench covers 10 major chronic disease categories, and each category contains samples from two disease subtypes. To provide more medically informative analysis, we further report method performance for each chronic disease subtype. These results offer a more fine-grained view of where different methods perform well or struggle across disease settings. The detailed results are presented in Tables 10 and 11.

The disease-wise breakdown shows that the advantage of stronger memory systems is broadly consistent rather than concentrated in a few categories. Letta achieves the best results in most disease groups, while MemOS and MemRL remain competitive in several categories such as chronic kidney, digestive, and endocrine diseases, indicating that the main findings are robust to clinical topic variation. At the same time, the gap between methods becomes especially

Table 10: Results by chronic disease category (Part I). Each value reports performance (%) under the corresponding disease category. The best result in each column is indicated in **bold**.

Baselines	Diabetes	Cardiovascular	Chronic Respiratory	Neurodegenerative	Chronic Kidney
Unstructured					
BM25-RAG	33.0	39.7	41.7	38.1	33.0
Embedding-RAG	44.2	46.2	46.2	47.7	33.0
Graph-Structured					
GraphRAG	29.9	46.2	45.4	47.4	31.4
HippoRAG-v2	33.5	33.7	33.7	37.6	30.5
ReMem	31.5	35.2	29.6	36.0	29.5
Flat-Extracted					
Mem0	37.6	39.7	36.2	36.0	34.0
LightMem	38.6	37.7	35.7	42.1	36.0
MemOS	50.8	54.3	51.3	54.3	52.0
MemRL	46.2	51.8	46.2	47.7	45.5
A-Mem	46.2	46.2	43.2	50.8	43.0
Multi-Compartment					
Letta	55.8	59.8	52.8	61.4	43.0
MIRIX	38.6	42.7	38.2	40.1	33.0

large in categories that require integrating long-term symptoms, treatments, and temporal progression, which further supports our claim that longitudinal medical memory demands structured cross-session reasoning rather than shallow lexical matching.

D Prompt Template

D.1 Evaluation Prompt Used During Benchmarking

MedMemoryBench uses a two-stage prompting pipeline during benchmarking: **Memory Construction** and **Query Answering**. In the prompt templates below, placeholders are written as `<field_name>` and are filled dynamically at runtime. For example, `<context>` refers to the current medical dialogue record, `<memory_source>` denotes the retrieved or accumulated memory used for answering, and `<question>` refers to the benchmark query.

Memory Construction. In the memory-construction stage, the model receives the current dialogue context and is asked to read and memorize the medically salient information. We use one shared template for this stage.

Memorize Prompt

Instruction: The following is a medical dialogue record. Please read it carefully and memorize the key information.

`<context>`

This prompt is intentionally lightweight. Its purpose is to encourage faithful retention of clinically important facts from the dialogue record without adding unnecessary reasoning constraints during memory writing.

Query Answering. In the query-answering stage, we first provide a shared system instruction to define the assistant role, and then pair it with a query-specific user prompt template. Because MedMemoryBench contains heterogeneous question types with different answer targets and reasoning demands, we use a dedicated template for each query category.

Table 11: Results by chronic disease category (Part II). Each value reports performance (%) under the corresponding disease category. The best result in each column is indicated in **bold**.

Baselines	Autoimmune	Digestive	Chronic Neuropsychiatric	Musculoskeletal	Endocrine
Unstructured					
BM25-RAG	36.4	31.7	33.3	30.2	42.0
Embedding-RAG	41.4	41.2	41.9	40.7	45.0
Graph-Structured					
GraphRAG	37.5	30.8	33.9	35.5	40.0
HippoRAG-v2	39.4	33.2	29.3	33.2	34.5
ReMem	33.4	28.8	29.2	31.7	36.5
Flat-Extracted					
Mem0	36.8	38.4	34.4	37.9	39.5
LightMem	37.9	42.2	33.8	40.7	43.0
MemOS	50.0	51.3	48.5	53.3	51.0
MemRL	52.5	50.3	46.5	48.2	56.0
A-Mem	47.5	45.2	42.4	46.2	44.0
Multi-Compartment					
Letta	55.1	51.3	55.1	76.5	55.4
MIRIX	40.4	37.7	33.3	44.7	45.0

Shared System Prompt

You are the patient’s personalized medical assistant, capable of accurately memorizing the patient’s complete medical history. Please reason and respond based on patient information in memory, maintain a warm yet professional tone, answer directly, and avoid unnecessarily long explanations.

EEM Answer Prompt

Context: Based on <memory_source>, accurately answer the following question.

Question: <question>

Answer Requirements:

1. Provide the target entity name directly.
2. Keep the answer brief and precise.
3. Do not include lengthy explanations.

Answer:

The **EEM** template is designed for precise slot-level retrieval. It therefore emphasizes direct extraction of the target entity rather than extended explanation.

TLA Answer Prompt

Context: Based on <memory_source>, accurately answer the following question.

Question: <question>

Answer Requirements:

1. If the question asks about a time, answer in YYYY-MM-DD format (e.g., 2024-01-15).
2. If the question asks about an event at a specific time, clearly describe the event content and key details.

3. Keep the answer concise and directly grounded in memory.

Answer:

The **TLA** template explicitly constrains temporal questions to normalized date outputs whenever possible, while still allowing concise event descriptions when the query asks what happened at a particular time point.

SUA Answer Prompt

Context: Based on <memory_source>, accurately answer the following question.

Question: <question>

Answer Requirements:

1. Describe the patient's most recent status.
2. Reflect important changes over time when necessary.
3. Maintain a warm yet professional tone.
4. Be concise and direct.

Answer:

The **SUA** prompt emphasizes up-to-date patient status and trajectory-aware summarization, which is important for questions that ask for the latest condition rather than isolated historical facts.

MQ Answer Prompt

Context: Based on <memory_source>, and considering the patient's allergy history, medical history, medications, and personal preferences, answer the following question.

Question: <question>

Answer Requirements:

1. Select all correct options.
2. Output only the option letter(s), such as B or B,D.
3. Do not provide any explanation.

Answer:

The **MQ** template enforces a strict multiple-choice output format, which simplifies automatic evaluation and avoids verbose justifications that are irrelevant to the benchmark target.

IG Answer Prompt

Context: Based on <memory_source>, and considering the patient's allergy history, medical history, medications, and personal preferences, answer the following question.

Question: <question>

Answer Requirements:

1. Reason from this patient's specific remembered information; do not give generic medical advice.
2. Maintain a warm yet professional tone.
3. Be concise, direct, and avoid boilerplate.
4. If recommending or advising against something, briefly explain the reason based on the patient's specific situation.

Answer:

The **IG** template is designed for personalized medical inference. It explicitly discourages generic recommendations and instead requires patient-grounded reasoning based on remembered allergy history, medication use, prior diagnoses, and personal preferences.

MCD Answer Prompt

Context: Based on <memory_source>, carefully review the patient’s complete medical history and conduct a comprehensive analysis by combining information from multiple visits.

Question: <question>

Answer Requirements:

1. Clearly list the memory content you draw upon.
2. Present a clear reasoning path from evidence to conclusions.
3. Provide a final comprehensive judgment.

Answer:

Finally, the **MCD** template targets queries that require multi-visit synthesis and explicit reasoning over dispersed historical evidence. Compared with the other templates, it places the strongest emphasis on transparent reasoning paths and comprehensive judgment grounded in multiple memory items.

D.2 LLM-as-Judge Prompt Template

For MedMemoryBench, we use LLM-as-Judge for four query types whose outputs cannot be reliably evaluated by exact string matching alone: **TLA**, **SUA**, **IG**, and **MCD**. These categories require judgment over temporal equivalence, state consistency, patient-grounded inference, or multi-hop reasoning quality. Accordingly, we design a dedicated judge prompt for each type so that the evaluator can apply query-specific criteria rather than a uniform matching rule.

More specifically, **TLA** requires verifying whether the model correctly identifies a time point or correctly describes the event associated with a queried time; **SUA** requires checking whether the answer reflects the patient’s *latest* status and whether it is grounded in remembered patient history rather than generic knowledge; **IG** requires evaluating whether the model reaches the right personalized recommendation or conclusion by explicitly using patient-specific evidence; and **MCD** requires a substantially stricter assessment of node coverage, causal-chain correctness, and reasoning completeness across multiple visits.

Below we present the prompt templates used for LLM-based judgment.

TLA Judge Prompt

You are a strict medical dialogue evaluation judge. Determine whether the model’s answer correctly addresses the time-related question.

Question:

<question>

Reference Answer:

<expected_answer>

Answer Explanation:

<explanation>

Model’s Answer:

<model_output>

Evaluation Criteria:

This is a temporal localization question, which may take one of the following two forms:

1. Asking when a certain event occurred. The model must correctly provide the time point
2. Asking what happened at a certain time. The model must correctly describe the event content

Judge strictly:

- If the model’s answer contains the correct time point or the correct event content, judge as [CORRECT]
- If the model’s answer about the time/event does not match the reference answer or fails to answer, judge as [INCORRECT]
- Date formats do not need to be identical, but must refer to the same time point (e.g., “January 1, 2024” and “2024-01-01” are considered equivalent)

Output in the following JSON format:

"is_correct": true/false, "reason": "brief justification"

Output JSON only, no other content.

The TLA judge prompt focuses on temporal equivalence. It allows surface-form variation in date expressions, but requires exact agreement at the level of the referenced time point or event content.

SUA Judge Prompt

You are a very strict medical dialogue evaluation judge. Determine whether the model’s answer correctly reflects the patient’s most recent status.

Question:

<question>

Reference Answer:

<expected_answer>

Answer Explanation:

<explanation>

Model’s Answer:

<model_output>

Evaluation Criteria:

This is a state update question, testing whether the model correctly answers the latest status based on the patient’s historical information in memory.

Core Evaluation Principles (critically important):

1. Must be based on memory: The model’s answer must demonstrate the use of the patient’s past memory information, not guessing or generic medical knowledge.
2. No guessing allowed: If the model has not retrieved relevant memory information but gives a “coincidentally correct” answer, it should be judged as [INCORRECT].
3. Information source requirement: A correct answer should convey that the model “remembers” this patient’s specific situation, rather than guessing.

Judge strictly:

- If the model’s answer demonstrates the use of patient historical memory and the core content is consistent with the reference answer, judge as [CORRECT]
- If the model’s answer contains key information points from the reference answer, and these clearly originate from patient memory retrieval, judge as [CORRECT]
- If the model’s answer clearly contradicts the reference answer, omits key information, or provides outdated status, judge as [INCORRECT]
- If the model states it does not know or cannot answer, judge as [INCORRECT]
- If the model’s answer appears too generic, lacks specific patient information support, or seems like a guess, even if the content happens to be close to the reference answer, judge as [INCORRECT]

Output in the following JSON format:

"is_correct": true/false, "reason": "brief justification, must indicate whether the model demonstrated use of patient memory"

Output JSON only, no other content.

The SUA judge prompt is designed to distinguish genuine memory-grounded answers from plausible but unsupported status summaries. In particular, it explicitly treats unsupported guessing as incorrect even when the final answer is superficially close to the reference.

IG Judge Prompt

You are a very strict medical dialogue evaluation judge. Determine whether the model’s reasoning answer is correct.

Question:

<question>

Reference Answer:

<expected_answer>

Answer Explanation:

<explanation>

<metadata_info>

Model's Answer:

<model_output>

Evaluation Criteria:

This is an inference generation question, testing whether the model can perform correct medical reasoning based on patient-specific information.

Core evaluation points:

1. Patient Information Utilization (Key)

- The model must demonstrate the use of patient-specific information from memory
- If required_patient_info is provided in metadata, the model's answer must reflect understanding of these key pieces of information (important)
- If the patient's specific circumstances and past memories are ignored or missing, judge as [INCORRECT]

2. Reasoning Quality

- The model must reason based on retrieved patient historical information, not purely from its own medical common sense
- If only a conclusion is given without sufficient reference to patient information and memory, judge as [INCORRECT]
- If the model gives a "common wrong answer" type of response (generic advice), judge as [INCORRECT]

3. Conclusion Correctness

- The final recommendation/conclusion should be fully consistent with the reference answer in direction
- Even if the conclusion is correct, if it lacks reasoning based on patient information, still judge as [INCORRECT]

Judgment rules:

- [CORRECT]: Answer uses patient-specific information, contains required patient information points, and reaches an accurate conclusion
- [INCORRECT]: Answer does not adequately consider the patient's specific circumstances
- [INCORRECT]: Answer ignores certain key information in required_patient_info
- [INCORRECT]: Answer matches the common_wrong_answer pattern
- [INCORRECT]: Model refuses to answer or claims no information

Output in the following JSON format:

"is_correct": true/false, "reason": "brief justification"

Output JSON only, no other content.

The **IG** judge prompt emphasizes patient-grounded reasoning rather than generic recommendation quality. This design is especially important because clinically plausible but non-personalized advice should not receive credit in MedMemoryBench.

MCD Judge Prompt

You are an extremely strict medical multi-hop reasoning evaluation judge. Your task is to rigorously verify whether the model truly retrieved and used specific information from the patient's historical memory to perform multi-hop clinical reasoning.

Question:

<question>

Reference Answer:

<expected_answer>

Answer Explanation:

<explanation>

<nodes_for_validation>

<required_nodes_str>

Reasoning Hops: <hop_count>
Reasoning Pattern: <reasoning_pattern>
Model's Answer:
<model_output>

--

Evaluation Task: Strict Node-by-Node Reasoning Chain Verification

This is a multi-hop clinical reasoning question. The core assessment is whether the model can accurately retrieve and use specific personalized medical information from the patient's historical memory.

Key Evaluation Principles (must be strictly followed)

- Patient-Specific Information Principle:** The model must explicitly reference the patient's specific data (such as specific test values, medication dosages, specific timing of symptom onset, particular diagnostic results), rather than giving generic medical common sense.
- 'Poor blood sugar control may lead to...'. This is generic medical knowledge, not patient-specific information
- 'Your fasting blood glucose rose from 6.8 to 8.2...'. This is patient-specific information
- Memory Retrieval Evidence Principle:** If the model fails to demonstrate specific references to the patient's historical records, even if the reasoning direction is correct, it should be judged as inadequate. The model must show it 'remembers' the patient's specific situation.
- Strict Causal Chain Correspondence Principle:** The causal relationships established by the model must precisely correspond to the causal mechanisms described in the reasoning chain nodes. Similar but different mechanisms cannot substitute.
- Node Content Precise Matching Principle:** During node verification, it is not sufficient to judge as 'covered' merely because the model mentioned a related concept. You must verify whether the model referenced the core specific content within the node.

Evaluation Steps

Step 1: Strict Node-by-Node Check

For each node in the reasoning chain, all of the following conditions must be verified:

Condition A - Specific Information Match:

- Did the model mention the specific data/time/event in this node?
- If the node contains specific values (e.g., 'TSH 0.02'), the model must mention the same or equivalent value
- If the node contains a specific time (e.g., 'October 2024'), the model must demonstrate awareness of that time point
- Merely mentioning the related concept (e.g., 'thyroid function') without specific data does not count as coverage

Condition B - Correct Causal Mechanism:

- Does the causal mechanism described by the model exactly match the reference reasoning chain?
- Using a different pathophysiological explanation (even if it sounds reasonable) does not count as correct
- Skipping intermediate steps to reach a conclusion directly does not count as correct

Condition C - Clear Information Source:

- Does the model's answer clearly demonstrate that this information comes from the patient's historical memory?
- Inferences based purely on medical common sense cannot receive credit

Step 2: Calculate Three Scoring Dimensions (Strict Standards)

NCR (Node Coverage Rate) = Number of nodes fully satisfying Condition A / Total number of nodes

- Mentioning concept only without specific data. Node not counted as covered
- Incorrect data or mismatched timeline. Node not counted as covered

CRC (Causal Relation Correctness) = Number of correctly established causal links / Number of expected causal links

- Must be the causal mechanism described in the reference answer; 'equivalent substitutions' not accepted


```

- Causal links skipping intermediate nodes. No credit
CC (Chain Completeness)
- 1.0 = Complete coverage of all nodes with correct causal relations
- 0.7 = Coverage of 80%+ nodes, core causal relations correct
- 0.5 = Coverage of 60%+ nodes, main causal relations correct
- 0.3 = Partial node coverage, causal relations have gaps
- 0.0 = No valid reasoning chain or completely wrong direction

Step 3: Comprehensive Judgment (High Standards)

[CORRECT] Conditions (all must be satisfied simultaneously):
- NCR  $\geq 0.75$  (at least three-quarters of nodes' specific content covered)
- CRC  $\geq 0.75$  (causal relations basically complete and correct)
- CC  $\geq 0.7$  (reasoning chain basically complete)
- Final conclusion consistent with reference answer

[PARTIALLY CORRECT] Conditions:
- NCR  $\geq 0.5$  and CRC  $\geq 0.5$  and CC  $\geq 0.5$ 
- Main reasoning direction correct, but with notable gaps

[INCORRECT] Conditions (any one triggers incorrect judgment):
- Model failed to retrieve patient-specific information from memory
- Reasoning based on generic medical knowledge rather than patient-specific situation
- Causal mechanism inconsistent with reference answer
- Conclusion direction incorrect
- NCR  $< 0.5$  or CRC  $< 0.5$  or CC  $< 0.5$ 

Output Format

Output your strict evaluation result in the following JSON format:
{
  "node_validations": [
    {
      "node_id": 1,
      "mentioned": true/false,
      "specific_data_matched": true/false,
      "causal_link_correct": true/false,
      "note": "Must state: 1) What specific data the model mentioned 2) Whether it precisely
matches node content 3) Whether the causal relation is correct"
    }
  ],
  "ncr_score": 0.0-1.0,
  "crc_score": 0.0-1.0,
  "cc_score": 0.0-1.0,
  "memory_retrieval_quality": "excellent/good/partial/poor/none",
  "uses_patient_specific_info": true/false,
  "is_correct": true/false,
  "reason": "Comprehensive justification, must state: 1) Whether the model used
patient-specific information 2) Which nodes were not covered 3) Whether the causal chain is
complete"
}

Output JSON only, no other content.

```

Among the four judge prompts, the **MCD** prompt is the most stringent. Instead of evaluating only the final answer, it requires explicit verification of patient-specific evidence usage, intermediate node coverage, and causal-chain faithfulness. This design makes the judge more sensitive to whether a model truly performs multi-hop reasoning over longitudinal medical memory rather than producing a plausible high-level summary.

E Special Implementation Details of the Baselines

For baseline implementation within the MedMemoryBench evaluation framework, we re-engineered most methods' HTTP clients, timeout handling, and TCP keepalive configurations to support more robust large-scale evaluation and

accurate accounting of API token consumption. For several baselines, additional method-specific adaptations were necessary due to constraints in their original implementations.

E.1 MIRIX

In the original MIRIX framework, the interactive dialogue mode relies on an internal MetaAgent to decide the retrieval strategy automatically. To ensure fair evaluation and precise token accounting, we instead use a manual mode: we explicitly call `retrieve_memory(memory_type='all')` to perform fused retrieval over all five memory types, namely Episodic, Semantic, Procedural, Resource, and Knowledge Vault, and then use an external LLM to generate the final answer. Because the original MIRIX retrieval mechanism is designed for a single memory type at a time, this all-type fused retrieval is a task-specific adaptation introduced in our evaluation framework.

E.2 HippoRAG-v2

HippoRAG-v2 was originally designed for one-shot indexing of a document collection followed by querying. Applying this design directly in our setting would introduce substantial time overhead. We therefore implement a cumulative-trigger scheme: for non-final sessions, raw text is buffered only into `_pending_sessions`; when `is_last_session=True`, we invoke `hipporag.index(docs)` once to build the knowledge graph in batch.

E.3 ReMem

Because each session in MedMemoryBench is relatively long, ReMem’s internal chunking pipeline would introduce excessive API calls, leading to substantial time and token costs. We therefore set `preprocess_chunk_func='none'` to disable ReMem’s internal text chunking. Each session is passed as a single document, and most sessions are thus processed without further splitting, substantially reducing the number of LLM calls.

F Ethical, Privacy, and Safety Considerations

Ethical review scope. MedMemoryBench is constructed for research on long-term medical memory rather than for clinical deployment. The benchmark data are generated and curated within a controlled pipeline and do not contain directly identifiable personal information. Even so, we treated the benchmark as a medical-domain resource with elevated risk and conducted internal review of data generation, annotation, storage, and release procedures before using it in experiments.

Privacy protection. Our construction pipeline is designed to avoid exposing personal identifiers at the source. During annotation and quality control, we restrict access to authorized staff, avoid recording unnecessary personal attributes, and focus review on medical consistency, answer correctness, and expression standardization. We do not release annotator identities, internal annotation records, or other operational metadata that could create secondary privacy risks.

Risk mitigation and intended use. We position MedMemoryBench as a research-only benchmark for evaluating memory retrieval and reasoning under longitudinal medical dialogue. It should not be used to support diagnosis, treatment recommendation, triage, or other real clinical decisions. We therefore recommend that future users keep human oversight in the loop, avoid deploying benchmark prompts or outputs directly in patient-facing systems, and perform separate ethics, privacy, and safety review if they extend the benchmark with real clinical data. Any future expansion toward real-world or multimodal medical records should additionally undergo institution-specific ethical review, stricter de-identification, and access control before public release.

G Limitation and Future Work

Synthetic-data bias remains unavoidable. Although we introduced human annotation for verification and correction, the raw dialogues and much of the underlying content are still generated by LLMs. As a result, the benchmark may retain model-specific priors and may not fully match the diversity of real patient–doctor interactions.

Scale and coverage are still limited. Due to annotation cost and budget constraints, the disease spectrum is not exhaustive and may be biased toward conditions that are easier to synthesize and standardize. In addition, despite its relatively large size, the dataset still contains limited numbers of sessions and patient trajectories.

The evaluation setting has limited ecological validity. Our benchmark mainly evaluates memory retrieval and reasoning in a controlled offline setting. It does not explicitly assess memory interpretability or editability, nor does it provide a deep evaluation of privacy and safety, which are important in medical applications.

The benchmark is text-only. MedMemoryBench does not yet cover multimodal medical memory, such as laboratory reports, medical images, prescriptions, or wearable signals, and therefore cannot fully reflect real-world clinical memory scenarios.

In future work, we plan to extend the benchmark toward multimodal medical data and to incorporate real user interaction data, so that it can better capture practical healthcare workflows and more realistic memory evolution patterns.